The Impact of Eccentric Phase Duration on Indices of Muscular Performance and Hypertrophy: A Systematic Review

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1.0 Introduction

Resistance training (RT) stands as a fundamental element in optimizing both strength and hypertrophy adaptations. These adaptations are influenced by numerous variables, including volume (Ralston et al., 2017; Schoenfeld, Ogborn, & Krieger, 2017), load (Lopez et al., 2021), proximity-to-failure (Refalo et al., 2023), and inter-set rest intervals (Grgic et al., 2017, 2018), among others. However, the impact of movement tempo (i.e., the speed at which repetitions and their respective phases are executed) has received relatively little attention.

In an early meta-analysis, Schoenfeld and colleagues revealed that extremely slow repetitions (>10 seconds per repetition) led to inferior muscle hypertrophy compared to faster repetition durations (0.5-8 seconds) (Schoenfeld et al., 2015). Subsequent reviews reported mixed results regarding strength and hypertrophy when comparing fast and moderately slow execution tempos (Davies et al., 2017; Hackett et al., 2018). Notably, these conclusions were based on a limited number of studies, and many of them combined various forms of tempo manipulation. As a result, it remains challenging to determine the distinct effects of altering specific contraction phases versus the overall repetition duration.

Emerging evidence suggests that the speed of individual contraction phases may hold more relevance for both strength and hypertrophy. For instance, a recent meta-analysis found greater strength gains when the concentric phase was performed rapidly (<2 seconds) compared to slower (≥2 seconds) (ES = 0.21, p = 0.029) (Hermes & Fry, 2023). This suggests that faster concentric contractions may be preferable for enhancing strength, potentially due to increased difficulty and diminished neuromuscular performance associated with slowing down the concentric phase, irrespective of overall repetition duration (Arazi et al., 2014; Lacerda et al., 2017).

Conversely, the impact of extending the eccentric phase has received less attention. Eccentric-only contractions are observed to result in hypertrophy of greater and similar magnitude than pure concentric contractions and combined eccentric-concentric contractions, respectively (Sato et al., 2022; Schoenfeld, Ogborn, Vigotsky, et al., 2017). While slowing down the eccentric phase has been linked to reduced resistance training performance (Headley et al., 2011; Pryor et al., 2011; Wilk et al., 2018, 2019; Wilk, Gepfert, et al., 2020; Wilk, Golas, et al., 2020), preliminary findings do not indicate that this acute performance decline negatively impacts long-term adaptations (Pereira et al., 2016). Consequently, it could be hypothesised that slowing down the eccentric phase while maintaining a faster concentric contraction may promote greater hypertrophy without impeding strength improvements.

To date, two reviews have qualitatively analysed the impact of slowing down the eccentric duration (Handford et al., 2022; Wilk et al., 2021), yielding differing conclusions. In addition, quality assessment of included studies was not performed. Given the inconsistency and gaps in the literature, a meta-analytic approach may be more appropriate to assess the impact of different eccentric tempos on strength and hypertrophy. Thus, we aim to conduct a systematic review with meta-analysis that seeks to quantitatively evaluate the impact of varying eccentric tempos on strength and muscle hypertrophy adaptations..

2.0 Methods

2.1 Overview and Registration

This systematic review is to be performed in accordance with the guidelines outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* (version 6.4.0) and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Page et al., 2021). The review was prospectively registered on the Open Science Framework after pilot searches had been performed, but before any formal searches or screen had been performed.

2.2 Search Strategy

A comprehensive search strategy using various terms related to resistance exercise and eccentric phase duration was developed in accordance with the PRISMA-S checklist (Rethlefsen et al., 2021), assisted by three specialized tools: the Word Frequency Analyzer (https://sr-accelerator.com/#/wordfreq), Research Refiner (https://sr-accelerator.com/#/searchrefinery), and Polyglot Search Translator (https://sr-accelerator.com/#/polyglot) (Clark, Glasziou, et al., 2020; Clark, Sanders, et al., 2020). The searches will be conducted in the following bibliographic databases, spanning from their inception to November 10th, 2023: PubMed/MEDLINE, SCOPUS, SPORTDiscus, CINAHL, ProQuest and Google Scholar. This comprehensive approach is designed to maximize the chances of locating both peer-reviewed and grey literature. No predefined limits or filters will be applied to the searches, although for Google Scholar, the search will be limited to the first 1,000 hits, and for ProQuest, searches will be limited to scholarly journals, dissertations and theses. The full search strings can be found Supplementary file I. Additionally, secondary searches will be performed, including: 1) Screening reference lists of all included studies and relevant review papers; 2) Forward citation tracking through Google Scholar for the studies included in the review; and 3) Ongoing search alerts will be set up after the initial search date until the week before submission.

2.3 Text Screening

After removal of duplicates, titles and abstract will be independently screened for initial eligibility by the two researchers (CA and AK) using the systematic review software Rayyan (https://rayyan.ai/). Disagreements will be resolved through discussion or in consultation with an outside reviewer when required. Search results from Google Scholar will be mass exported using the *Publish or Perish* software (Harzing, 2007).

2.4 Inclusion and Exclusion Criteria

The current systematic review will include studies that investigated the impact eccentric phase duration as the independent variable, which will be classified either fast (i.e., ≤2 sec) or slow (i.e., >2 sec or more or completed purposely slower than fast or volitional tempo), and (a) was published in a peer-reviewed journal, on a pre-print repository, or as a Master's or PhD thesis, (b) are written in the English language, (c) include healthy human participants with no disease or musculoskeletal injury, (d) used a parallel groups or contralateral, within-subjects design to assess the effects of altered eccentric phase duration on indices of muscle strength, power, endurance, and/or hypertrophy, (e) used traditional combined eccentric/concentric isotonic contractions, (f) the study matched the duration of all contraction phases except the eccentric (i.e. concentric and possible isometric transition phases), (g) the study matched the frequency, relative-load (% of 1RM or RM relative to tempo) and volume (number of sets or volume-load) between groups, and (h) had an intervention period of at least 4 weeks. Studies employing eccentric-only and overloaded eccentrics were excluded.

2.5 Study Coding and Extraction

The following data will be extracted from the included studies: 1) number of participants and descriptives (including age, sex, body mass, height, and training experience); 2) resistance training protocol including duration, frequency, volume, relative-load, rest periods, and exercise selection; 3) study design; 4) tempo of each contraction phase; 5) outcome measures; 6) the presence of concurrent training; and 7) change means and standard deviations of the relevant indices of muscular performance and hypertrophy. Where insufficient data is reported, the corresponding author will be contacted by email twice over two months. If they fail to provide the required data, data will be extracted using WebPlotDigitizer and/or calculated in accordance with the Cochrane Handbook (Higgins et al., 2019). All data extraction will be completed by CA and coding files will be cross-checked by AK for accuracy and differences will be resolved through discussion and consensus.

2.6 Risk of Bias

Risk of bias assessment will be performed using the second version of the Cochrane RoB tool for randomized trials (RoB 2) (Sterne et al., 2019), with signalling questions 2.1 and 2.7 removed due to the inability to blind participants in RT-trials and the effect of adhering, not assignment, to the protocol is of interest. Studies will be categorized as follows: "Low risk of bias" if all domains are assessed as low risk; "Some concerns" if concerns are raised in one domain, but no domain is determined to be high risk; "High risk of bias" if at least one domain is classified as high risk or if multiple domains raise some concern. This will be performed independently by two researchers (CA

and AK). Any disagreements will be resolved through discussion or, if necessary, by consulting a third reviewer. The Grading of Recommendations Assessment, Development and Evaluation (GRADE) system will be used to evaluate the certainty of evidence for the studies included in the quantitative synthesis (Atkins et al., 2004), in a similar manner to a previous exercise science review (King et al., 2022).

2.7 Meta-Analysis

Meta-analysis will be performed - if possible, depending on the number of studies and their heterogeneity – on indices of muscular strength, power, endurance and hypertrophy. For hypertrophy, both direct (cross-sectional area, muscle thickness and volume etc.) and indirect (fat free mass) will be included, but analysed separately, and only if sufficient studies are located. Otherwise, the data will be qualitatively interpreted. Additionally, sub-group analyses will be performed on the effects of study design (i.e., within- or between-subject design), participant training status (i.e., trained or untrained), body part (upper or lower body) and the magnitude of eccentric phase enhancement (i.e., slow = 3-5 seconds or very slow = >5 seconds).

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